Optimizing Urban Mobility: A Review of Hybrid Smart Parking Systems Integrating Blockchain, Deep Prediction, and Heuristic Control

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***Abstract*—The optimization of urban parking resources is a multi-dimensional challenge requiring the convergence of accurate forecasting, secure transactions, and efficient resource allocation. This paper presents a systematic review of 25 recent studies (2015–2025) on Integrated Smart Parking Systems (SPS). We identify a critical architectural dichotomy: while Deep Learning models (LSTM, Graph-GRU) have become the standard for *occupancy prediction*, they are often decoupled from the *control layer*. The literature favors deterministic approaches—specifically Stackelberg Games and Evolutionary Algorithms—over Deep Reinforcement Learning (DRL) for decision-making due to the latter's sample inefficiency. Furthermore, we analyze the role of Fog Computing and Blockchain in decentralized infrastructure. We conclude that future frameworks must move toward "Hybrid Intelligence," combining deep predictive power with heuristic control logic.**

***Index Terms*—Smart Parking, Blockchain, Deep Learning, Heuristic Optimization, Game Theory, IoT, Fog Computing.**

I. Introduction

The rapid acceleration of global urbanization presents one of the most significant challenges to modern city infrastructure. With approximately 80% of the world's population projected to reside in urban centers by 2050 [16], the proliferation of vehicles has outpaced the development of supporting infrastructure. A critical consequence of this asymmetry is the phenomenon of "cruising"—vehicles circling continuously in search of vacant parking. Research indicates that this search traffic accounts for 30% of total urban congestion, spiking to nearly 74% in downtown areas during peak hours [5], [16]. This inefficiency is not merely a logistical nuisance but a severe economic and environmental hemorrhage. In the United States alone, drivers spend an average of 17 hours annually searching for parking—rising to 107 hours in high-density cities like New York—resulting in an estimated economic loss of $72.7 billion in wasted time and fuel [5], [16].

The environmental implications of this inefficiency are equally profound. A single district in Los Angeles wastes approximately 47,000 gallons of gasoline annually solely due to parking searches [5]. Furthermore, the vehicular dynamics of cruising exacerbate the carbon footprint; studies demonstrate that carbon monoxide and nitrogen hydride emissions increase by over 50% when vehicle speeds drop from 50 km/h to 20 km/h during the search process [16]. Consequently, the optimization of parking resource allocation is no longer a matter of convenience, but a prerequisite for sustainable urban ecology.

While Intelligent Transportation Systems (ITS) and Internet of Things (IoT) solutions have been proposed to mitigate these issues, they face significant deployment barriers, primarily regarding data scarcity and spatial modeling. Efficient prediction relies on dense sensor networks; however, installing real-time sensors on a city-wide scale is economically prohibitive. For instance, in major metropolises like Beijing, only approximately 6% of parking lots are equipped with real-time availability sensors [20]. This data scarcity renders traditional prediction models ineffective, as they fail to capture the complex, non-Euclidean spatial autocorrelations where the occupancy of one lot influences distant nodes within the traffic network [20].

Furthermore, existing Smart Parking Systems (SPS) exhibit critical vulnerabilities regarding security and user privacy. The predominant architecture of current SPS is centralized, relying on a single server to manage queries and reservations. This centralization introduces a Single Point of Failure (SPoF), making the infrastructure susceptible to Distributed Denial of Service (DDoS) attacks and remote hijacking [22]. More alarmingly, these systems often require users to disclose sensitive data—such as real identities, immediate destinations, and dwell times—to a central authority. Aggregating this data exposes users to severe privacy risks, allowing malicious actors or service managers to infer life patterns, work addresses, and income levels [22]. Despite these risks, few existing frameworks simultaneously address the trilemma of prediction accuracy under sparse data, system resilience, and user anonymity.

This paper provides a taxonomy of these approaches (see Fig. 1) and identifies the gap between high-accuracy Deep Learning and low-latency Edge Computing.

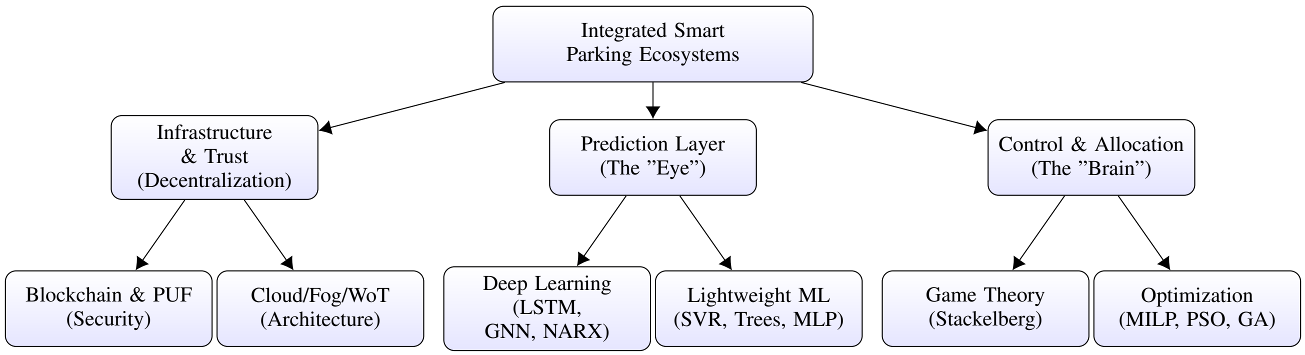

Fig. 1. Taxonomy of approaches identified in the reviewed literature.

II. Infrastructure: IoT, Fog, and Blockchain

The foundation of modern SPS is the shift from centralized servers to decentralized and distributed computing.

A. Cloud, Fog, and Web of Things

To handle the massive influx of sensor data, Razzaq et al. [4] proposed a "Cloud of Things" (CoT) framework, utilizing geofences to trigger vehicle-to-infrastructure (V2I) communication only when necessary. However, cloud latency remains an issue. Enríquez et al. [23] addressed this by implementing a Fog Computing architecture, processing data closer to the edge to minimize response delays. Similarly, Provoost et al. [12] leveraged the "Web of Things" (WoT) standard to integrate heterogeneous data sources, such as weather and traffic flows, into a unified semantic web. On a cooperative level, Alarbi et al. [19] introduced SCOPE, a distributed architecture that groups parking agents via geohashing to facilitate autonomous interaction.

B. Blockchain Security and P2P Economy

Decentralization also applies to trust. Badr et al. [22] and Turki et al. [17] focused on security, utilizing anonymous reputation management and Physical Unclonable Functions (PUF) to prevent sensor cloning. Economically, Hasan et al. [3] and Brenner et al. [5] utilized Ethereum and Hyperledger blockchains to enable Peer-to-Peer (P2P) parking sharing, allowing private owners to lease spots via smart contracts without intermediaries.

III. Prediction: Accuracy vs. Efficiency

A. Deep Learning Architectures

For high-fidelity forecasting, Deep Learning dominates. Paidi et al. [21] and Jin et al. [25] demonstrated that Long Short-Term Memory (LSTM) networks excel at handling non-linear dependencies. Sebatli et al. [16] compared statistical models against NARX neural networks, finding that while ARIMA fails in high-fluctuation zones, NARX maintains stability. Spatially, Feng et al. [13] and Zhang et al. [20] utilized Graph Neural Networks (GNN) to model zone-wise correlations, proving that parking availability is a graph-structured problem.

B. Lightweight and Vision-Based Models

For edge devices, lighter models are required. Bock et al. [1] and Fan et al. [14] utilized Support Vector Regression (SVR) with noise-filtering optimization. Bhanupriya et al. [24] and Zheng et al. [11] showed that ensemble methods like CatBoost and Regression Trees can outperform heavier neural networks in specific scenarios. Hardware-specific optimization was explored by Ismail et al. [15], who deployed a lightweight MLP on FPGA. Additionally, Rani et al. [10] utilized YOLO for real-time license plate recognition, providing the raw occupancy data for these models.

IV. Control Layer: Deterministic Optimization

A. Game Theory and Multi-Agent Systems

Allocating resources requires strategic interaction. Saharan et al. [6] used Stackelberg Games to model the leader-follower dynamic between controllers and drivers. Jioudi et al. [7] expanded this into a multi-agent framework, ensuring fair sharing of public space through dynamic negotiation.

B. Heuristic and Mathematical Optimization

Instead of "learning" policies via RL, many systems simply "solve" for the best outcome. Kotb et al. [8] used Mixed-Integer Linear Programming (MILP) to minimize monetary costs. Piyush et al. [9] combined Feedforward Neural Networks (FNN) for pathfinding with MILP for allocation. Evolutionary algorithms are also prominent; Sahu et al. [2] employed Particle Swarm Optimization (PSO) within a Digital Twin, while Abo-Zahhad et al. [18] utilized Genetic Algorithms (GA) to optimize self-parking maneuvers and steering control.

V. Conclusion and Future Research

The synthesis of 25 studies indicates that the next generation of Smart Parking Systems will rely on "Hybrid Intelligence," combining the predictive power of Deep Learning with the strategic decision-making of Game Theory and Optimization. While current systems have successfully reduced search times and energy consumption, several avenues for high-impact research remain open.

A. From Grid-Based to Graph-Based Prediction

Most existing prediction models treat city sectors as isolated grids. However, urban traffic is inherently interconnected and non-Euclidean. Future research should prioritize **Graph Neural Networks (GNNs)**, specifically Spatio-Temporal Graph GRUs (ST-GBGRU), to capture the spatial dependencies between distant parking zones [13]. By modeling parking lots as nodes in a graph, systems can predict how congestion in one area (e.g., a stadium) propagates to affect availability in residential zones miles away.

B. Digital Twins and Pareto Optimization

Current systems are largely reactive. The integration of **Digital Twin** technology—creating real-time virtual replicas of physical infrastructure—offers a shift toward proactive simulation. Future frameworks should utilize Digital Twins to run multi-objective scenarios using **Pareto Front Optimization** [2]. This allows city planners to visualize trade-offs in real-time, balancing conflicting objectives such as maximizing revenue versus minimizing carbon emissions or search time, before implementing policy changes.

C. Game Theoretic Dynamic Pricing

Static or simple demand-based pricing is insufficient for diverse user groups (e.g., delivery trucks vs. residents vs. shoppers). Future control layers should implement **Stackelberg Games** to model the negotiation between Parking Controllers (leaders) and drivers (followers) [6]. This introduces a social layer to the technical infrastructure, using dynamic pricing not just for revenue, but as a tool to incentivize drivers to park in underutilized areas, effectively ”gamifying” traffic distribution to reduce congestion.

D. Edge Intelligence and Privacy

Finally, as sensor density increases, cloud latency becomes a bottleneck. Research must pivot toward **Edge and Fog Computing**, where lightweight models (like optimized MLPs) run directly on local hardware [15], [23]. This not only reduces response time for autonomous vehicles but also enhances privacy by processing sensitive user data locally rather than transmitting it to a central server.

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TABLE I: Comparative Analysis of Literature on Smart Parking Systems

No.	Author (Year)	Methodology	Key Points	Conclusion/Outcome
1	Bock et al. (2017) [1]	2-Step SVR	Trend extraction via SVR smoothing followed by prediction.	Filters noise to leverage extended history; compresses storage by 60%.
2	Sahu et al. (2025) [2]	MDP & PSO	Digital Twin; Pareto Front Optimization; Markov Decision Process.	25% reduced search time; 18% energy savings; hybrid approach enables scalability.
3	Hasan et al. (2025) [3]	Blockchain Pricing	MySQL assisted Ethereum system; Dynamic weighted pricing.	Enhanced security; reduced search time via recommendations; user-friendly interface.
4	Razzaq et al. (2020) [4]	Cloud of Things	Geofence-Triggered Prediction; V2I communication.	Dynamically updates status based on real-time traffic logs; integrates RFID/WSN.
5	Brenner et al. (2023) [5]	DPark (Blockchain)	Smart contracts (Ethereum/Hyperledger); Benchmarked latency.	Eliminates central server vulnerabilities; examines blockchain workload impacts.
6	Saharan et al. (2023) [6]	Game Theory	Stackelberg games for pricing; Multi-Level Queue for priority.	Highest revenue ($381k) and acceptance rates; Fog architecture reduced overhead.
7	Jioudi et al. (2019) [7]	Multi-agent System	e-Parking platform; Multi-agent framework for dynamic pricing.	Improves driver experience via real-time pricing and reservation sharing.
8	Kotb et al. (2016) [8]	iParker (MILP)	MILP for minimizing cost/maximizing utilization.	Combines real-time and share-time reservations; offers choices on destinations.
9	Piyush et al. (2017) [9]	FNN & MILP	FNN for path processing; MILP for resource allocation.	Suggested pricing policies for static/reservations that maximize profit.
10	Rani et al. (2024) [10]	YOLO & LPR	Automatic License Plate Recognition (LPR) with YOLO.	¿90% recognition rate; significant improvement over traditional methods.
11	Zheng et al. (2015) [11]	Regression Trees	Compared RT, NN, SVR on SF/Melbourne data.	Regression Trees offered highest accuracy ($R^2 > 0.94$) and lowest cost.
12	Provoost et al. (2020) [12]	WoT & ML	Web of Things sensors; Comparison of NN, CNN, and RF.	ML models (MSE 7.18) outperformed state-of-the-art; integrated weather data.
13	Feng et al. (2025) [13]	Graph-GRU	Graph Convolution (GCN) embedded in GRU gates.	Fusing GCN into GRU yields superior accuracy; direct prediction minimizes error.
14	Fan et al. (2018) [14]	SVR & FOA	SVR optimized by Fruit Fly Optimization Algorithm (FOA).	FOA-SVR superior to BPNN/ELM in accuracy; automates parameter tuning.
15	Ismail et al. (2021) [15]	MLP on FPGA	Optimized MLP for FPGA edge devices.	16-hidden-unit MLP beat ARIMA; viable for low-power edge computing.
16	Sebatli et al. (2023) [16]	NARX vs ARIMA	Benchmarked ARIMA vs NARX Neural Networks.	NARX NN significantly outperformed ARIMA in high-fluctuation zones.
17	Turki et al. (2024) [17]	PUF & Blockchain	Physical Unclonable Functions (PUFs) for IoT authentication.	Enhances reliability/security against cloning and unauthorized access.
18	Abo-Zahhad et al. (2025) [18]	GA & SVM	Genetic Algorithms for control; SVM for anomaly detection.	High precision in occupancy tracking; scalable cloud management.
19	Alarbi et al. (2023) [19]	SCOPE	Cooperative distributed system; Geo-clustering.	Comprehensive solution for agent cooperation in open environments.
20	Zhang et al. (2022) [20]	Graph NN	Hierarchical Graph Convolution; Semi-supervised learning.	Outperforms baselines in data-scarce scenarios; high spatial robustness.
21	Paidi et al. (2022) [21]	LSTM vs SARIMAX	Open lot prediction; Google trends integration.	LSTM robust for sequential variances; SARIMAX failed on abrupt changes.
22	Badr et al. (2020) [22]	Blockchain Privacy	Privacy Information Retrieval (PIR); Reputation management.	Robust security with low overhead; prevents tracking and unfair pricing.
23	Enríquez et al. (2024) [23]	Fog Computing	Fog architecture with AdaBoost; Mobile app for user input.	TAdaBoost outperformed others; Fog modules reduced response delay.
24	Bhanupriya et al. (2025) [24]	CatBoost	CatBoost classifier compared to RF and Logistic Regression.	CatBoost yielded superior accuracy; effective at handling categorical features.
25	Jin et al. (2022) [25]	Improved LSTM	LSTM with weather vectors integrated into gating mechanisms.	Weather-enhanced LSTM achieved lowest error rates vs SVR/MLR.